IYFV 1.03 INTERMAGNET FORMAT FOR YEARMEAN FILE

Annual means of magnetic data with 1 nT or 0.1 min-of-arc resolution are organized on a year file basis. One file contains all annual mean values of the geomagnetic field components that are available from the observatory.

The file includes a header block, a data block and a footer block. The data block must have from 1 to 3 tables containing annual mean values. A table of annual means for ALL DAYS for each year of INTERMAGNET membership is required. Optionally, ALL DAYS annual mean data available prior to INTERMAGNET membership and tables of annual mean data for QUIET-DAYS and DISTURBED-DAYS are encouraged where possible.
The first table in the data block must contain ALL DAYS means, optionally followed by tables containing QUIET DAYS means and DISTURBED DAYS means in that order if these data are available. Each table must have at least one blank line at the end of the table (CrLf end-of lines, or 73 space characters followed by CrLF end-of-lines)
If DISTURBED DAYS means are available but QUIET DAYS means are not available then a table of QUIET DAYS means must be included between the ALL DAYS and DISTURBED DAYS tables with QUIET DAYS data filled with No Data Values.

File name:

"YEARMEAN" and the three-letter observatory ID code as an extension. eg: YEARMEAN.BOU for Boulder.

Description of the header block

The header contains information on file format version, observatory name, IAGA three-letter code, colatitude, longitude and elevation. It further contains the headers for each data column. Location data must be decimal degrees with three decimal places for colatitude and east longitude on the WGS-84 datum and elevation as integer metres above sea level.
The file format version information is identified by including “IYFV” as the first four characters in the file followed by a numerical format descriptor defining the format version. If “IYFV” is not include or the numerical format descriptor is not included or not defined as a format version then the file is interpreted as IYFV1.02 or earlier.

The header must follow the layout in the example shown below with each line no longer than 75 characters including CrLf end-of-lines.
eg: The header for an IYFV1.03 format yearmean data file for Wingst is:

```
IYFV1.03                    ANNUAL MEAN VALUES

                          WINGST, WNG, GERMANY

  COLATITUDE:  36.275      LONGITUDE:   9.053 E      ELEVATION:   66 m

   YEAR      D        I         H      X      Y      Z      F  * ELE Note
          Deg. min Deg. min     nT     nT     nT     nT     nT
```

Description of data block

Each line in the data block is 75 characters per line including CrLf. All data fields are right-justified. The field width must be maintained, either by zero-filling or space filling for number fields and space filling for character fields.
The '+'sign for positive values is optional.

```
_YYYY.yyy_DDD_dd.d_III_ii.i_HHHHHH_XXXXXX_YYYYYY_ZZZZZZ_FFFFFF_A_EEEE_NNNCrLf
....
....
```

YYYY.yyy Epoch is given with 3 decimals
DDD_dd.d Declination is given in degrees and decimal minutes of arc

III_ii.i	Inclination is given in degrees and decimal minutes of arc
HHHHHH	H-component is given in nT
XXXXXX	X-component is given in nT
YYYYYY	Y-component is given in nT
ZZZZZZ	Z-component is given in nT
FFFFFF	F-component is given in nT calculated from vector data; if F data from previous years were calculated using a continuously recording scalar magnetometer this should be identified in an explanatory note in the footer block.
A	Type of annual means
EEEE	Vector variometer sensor orientation eg “XYZF”, “HDZF”, “DFI”, “ABZ”
NNN	Note number
_	represents a space character
CrLf	indicates a Carriage return and Line feed character

Angular values of the magnetic field are expressed as degrees and minutes. Values may be written in the range 0 to 360 or -180 to 180. Observatories may choose which range to use. Negative values must always have the minus sign before the degree field, never before the minute field (including values between 0 and -1 degrees, for example "-0 59" means a value of minus zero degrees fifty nine minutes). This applies to all types of records, including jump records

A
The type of annual means may be "A"ll, "Q"uiet, "D"isturbed, "I"ncomplete" or "J"ump.

Any annual means calculated using greater than or equal to 50% but less than 90% of a full year of minute data must be identified as incomplete with “I”. Optionally, these data can be included as No Data Values and identified with “I”**.** Where there are less than 50% of a full year of minute data available the No Data Value must be used and the record identified as incomplete with “I”.
Where a data record is marked “I” it must be clarified with an explanatory note in the footer block.

“Q” means are calculated using minute data from the 5 international most quiet days for each month of the year as published by the International Service of Geomagnetic Indices.

“D” means are calculated using minute data from the 5 international most disturbed days for each month of the year as published by the International Service of Geomagnetic Indices.

"J" is not an annual mean value, but indicates a jump in the observatory recordings, the cause of which must be described in an explanatory note. Jump records contain measured differences between an old setup or location and a new set-up or location within an observatory data series. It is common practice to carry a jump forward or backward to the start of a year. Jump records should be repeated in every data table included in the data block (A, Q, D).

To update values (be they annual, monthly, hourly etc) before the jump so that they are consistent with values after the jump, the jump values should be subtracted. If the measured differences are in D (in degrees), H and Z (in nT) then those for I (in degrees), X, Y and F (in nT) are

$$\Delta I = \frac{180}{\pi}\frac{H\Delta Z - Z\Delta H}{F^2}$$

$$\Delta X = \Delta H \cos D - \frac{\pi}{180}\Delta \mathrm{DH} \sin D$$

$$\Delta Y = \Delta H \sin D + \frac{\pi}{180}\Delta \mathrm{DH} \cos D$$

$$\Delta F = \frac{H\Delta H + Z\Delta Z}{F}$$

And if the measured differences are in X, Y and Z (in nT) those in D and I (in degrees), H and F (in nT) are

$$\Delta D = \frac{180}{\pi}\frac{X\Delta Y - Y\Delta X}{H^2}$$

$$\Delta I = \frac{180}{\pi}\frac{H\Delta Z - Z\Delta H}{F^2}$$

$$\Delta H = \frac{X\Delta X + Y\Delta Y}{H}$$

$$\Delta F = \frac{X\Delta X + Y\Delta Y + Z\Delta Z}{F}$$

Where the full-field values are averages of the annual means before (uncorrected) and after the jump.

EEEE

Vector variometer sensor orientation. Two common examples are XYZF and HDZF. If an independent total field measurement is not made at an observatory this field should not include an 'F' code. For example, an observatory using a three component fluxgate with one horizontal sensor aligned along the magnetic meridian and a proton magnetometer should use 'HDZF' in this field. An observatory using only the fluxgate should use 'HDZ'. An observatory using a dIdD variometer should use “DIF”

YYYY.yyy

Epoch of the annual mean. Where ALL DAYS annual means are calculated from 90% or more of a full year of minute data the epoch should be listed as the middle of the year (for example 2022.500).

If there is less than 50% of a full year of minute data and an annual mean data record is included using No Data Values then the epoch is listed as the middle of the year, for example 2022.500.

If ALL DAYS annual means are calculated from greater than or equal to 50% but less than 90% of a full year of minute data and data are not represented by No Data Values the epoch must be listed as the average epoch of the available data, for example 2022.440.

QUIET DAYS and DISTURBED DAYS tables, if included, should use the same epochs as the corresponding ALL DAYS table.

No Data Value

Where data are not available for a particular year or element the data are replaced with a No Data Value. For the angular elements the No Data Value is “999 99.9”. For the magnitude elements, the No Data Value is “999999”.

For any years where no data are available or the No Data Value is used for all data fields, it is acceptable to omit that record entirely, except in the special case when a DISTURBED DAYS data table is included without any QUIET DAYS data table, as described above.

Description of the footer block

At the end of the file is added a footer describing the data. The footer contains notes on jumps, incomplete data sets etc. The footer must look like the example below with notes adjusted for each particular observatory. Each line in the notes section should be terminated with CrLF.:

- A = All days
- Q = Quiet Days
- D = Disturbed Days
- I = Incomplete
- J = Jump: jump value = old site value - new site value

ELE = vector variometer sensor orientation

Notes:

1. Note A
2. Note B

Example of a yearmean file

```
IYFV1.03                    ANNUAL MEAN VALUES

                      OBSERVATORY NAME, COD, COUNTRY

  COLATITUDE:  23.510      LONGITUDE: 333.863 E      ELEVATION:  4 meters

    YEAR       D         I         H      X      Y      Z      F  * ELE Note
           Deg.  '  Deg.  '      nT     nT     nT     nT     nT

 1942.500 -27 47.9  76 42.1  12973  11476  -6050  54889  56401 A  XYZ
 1949.500 -28 03.0 999 99.9  12946  11426  -6088 999999 999999 I  XYZ   1
 1950.650 -28 13.6  76 43.5  12940  11401  -6120  54843  56349 I HDZF   2
 1951.500 -28 22.9 999 99.9  12926  11372  -6144  54810  56313 A HDZF   3
 1983.568 326 41.6  77 15.8  12152  10156  -6673  53764  55120 I  DHZ   3
 1984.500 326 55.7  77 14.3  12171  10199  -6642  53736  55097 A  DHZ
 1985.500 327 11.1  77 12.9  12187  10242  -6604  53706  55071 A  DHZ
 1986.500 327 26.8  77 11.7  12201  10284  -6565  53679  55048 A  DHZ
 1987.500 327 44.5  77 09.9  12223  10336  -6524  53647  55022 A  DHZ
 1988.500 328 00.5  77 09.0  12235  10377  -6482  53633  55011 A  DHZ
 1989.000   0 02.6   0 00.7     -4      2     10     30     28 J  DHZ   4
 1989.500 328 13.8  77 07.2  12254  10418  -6452  53592  54975 A  DHZ
 1990.500 328 29.9  77 05.9  12271  10463  -6412  53571  54959 A  DHZ
 1991.500 328 45.6  77 04.9  12284  10503  -6371  53555  54946 A UVZF   5
 1992.500 329 01.3  77 03.4  12302  10547  -6332  53525  54920 A  DHZ
 1993.500 329 17.9  77 01.6  12323  10596  -6292  53495  54896 A  DHZ
 1994.000   0 00.0   0 00.0     -1     -1      0     -2     -3 J  DHZ   6
 1994.500 329 34.3  77 00.7  12335  10636  -6247  53476  54880 A  DHZ
 1995.500 329 53.6  76 58.3  12366  10698  -6203  53444  54856 A  DHZ
 1996.500 330 13.6  76 56.0  12395  10759  -6155  53409  54828 A  DHZ
 1997.500 330 33.9  76 54.0  12423  10819  -6105  53381  54807 A  DHZ
 1998.500 330 55.6  76 52.2  12446  10878  -6048  53361  54793 A  DHZ
 1999.500 331 17.3  76 50.2  12473  10939  -5992  53332  54771 A  DHZ
 2000.500 331 39.0  76 48.4  12497  10998  -5934  53311  54756 A  DHZ
 2001.500 332 01.3  76 46.1  12527  11063  -5877  53278  54731 A  DHZ
 2002.500 332 23.6  76 44.2  12553  11124  -5817  53254  54714 A  DHZ
 2003.500 332 45.2  76 43.3  12564  11170  -5752  53237  54699 A  DHZ
 2004.500 333 07.8  76 40.5  12600  11240  -5695  53202  54674 A  DHZ
 2005.500 333 29.3  76 38.7  12624  11296  -5635  53176  54654 A  DHZ
 2006.500 333 50.4  76 36.2  12656  11360  -5580  53140  54626 A  DHZ

 1983.568 326 42.3  77 15.1  12164  10167  -6677  53765  55124 I  DHZ   3
 1984.500 326 56.3  77 13.3  12186  10213  -6648  53734  55098 Q  DHZ
 1985.500 327 11.6  77 12.0  12202  10256  -6611  53704  55073 Q  DHZ
 1986.500 327 27.4  77 10.8  12215  10297  -6571  53676  55048 Q  DHZ
 1987.500 327 44.9  77 09.4  12232  10345  -6527  53648  55025 Q  DHZ
 1988.500 328 00.8  77 08.2  12246  10387  -6487  53631  55011 Q  DHZ
 1989.000   0 02.6   0 00.7     -4      2     10     30     28 J  DHZ   4
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1989.500	328	14.4	77	06.6	12263	10427	-6455	53591	54976	Q	DHZ	
1990.500	328	30.0	77	05.3	12279	10470	-6416	53567	54956	Q	DHZ	
1991.500	328	46.1	77	04.0	12297	10515	-6376	53551	54945	Q	UVZF	5
1992.500	329	01.6	77	02.7	12312	10556	-6336	53521	54919	Q	DHZ	
1993.500	329	18.2	77	00.9	12335	10607	-6297	53491	54895	Q	DHZ	
1994.000	0	00.0	0	00.0	-1	-1	0	-2	-3	J	DHZ	6
1994.500	329	35.4	76	59.2	12357	10657	-6255	53470	54879	Q	DHZ	
1995.500	329	54.2	76	57.5	12380	10711	-6208	53443	54858	Q	DHZ	
1996.500	330	13.6	76	55.5	12403	10766	-6159	53407	54828	Q	DHZ	
1997.500	330	34.2	76	53.4	12431	10827	-6108	53380	54808	Q	DHZ	
1998.500	330	55.5	76	51.6	12456	10886	-6053	53359	54793	Q	DHZ	
1999.500	331	17.9	76	49.6	12483	10949	-5995	53330	54771	Q	DHZ	
2000.500	331	39.3	76	47.8	12507	11007	-5938	53308	54755	Q	DHZ	
2001.500	332	01.5	76	45.6	12535	11070	-5880	53278	54733	Q	DHZ	
2002.500	332	23.7	76	43.6	12562	11132	-5821	53252	54714	Q	DHZ	
2003.500	332	45.9	76	42.0	12584	11189	-5759	53234	54701	Q	DHZ	
2004.500	333	08.1	76	39.7	12613	11252	-5700	53200	54675	Q	DHZ	
2005.500	333	29.6	76	37.8	12640	11311	-5641	53177	54659	Q	DHZ	
2006.500	333	50.5	76	35.5	12669	11371	-5585	53141	54630	Q	DHZ	
1983.568	326	40.4	77	17.7	12121	10128	-6659	53763	55112	I	DHZ	3
1984.500	326	54.6	77	16.5	12136	10168	-6626	53744	55097	D	DHZ	
1985.500	327	10.1	77	14.7	12158	10216	-6592	53707	55066	D	DHZ	
1986.500	327	25.6	77	13.7	12169	10255	-6552	53683	55045	D	DHZ	
1987.500	327	43.9	77	11.0	12205	10320	-6516	53645	55016	D	DHZ	
1988.500	327	59.5	77	10.9	12204	10349	-6469	53636	55007	D	DHZ	
1989.000	0	02.6	0	00.7	-4	2	10	30	28	J	DHZ	4
1989.500	328	12.2	77	08.9	12228	10393	-6443	53598	54975	D	DHZ	
1990.500	328	30.0	77	07.3	12249	10444	-6400	53577	54959	D	DHZ	
1991.500	328	45.1	77	06.5	12258	10480	-6359	53560	54945	D	UVZF	5
1992.500	329	00.8	77	05.6	12268	10517	-6316	53539	54927	D	DHZ	
1993.500	329	16.8	77	03.5	12295	10570	-6281	53502	54897	D	DHZ	
1994.000	0	00.0	00	00.0	-1	-1	0	-2	-3	J	DHZ	6
1994.500	329	33.2	77	02.9	12300	10604	-6233	53481	54877	D	DHZ	
1995.500	329	52.6	76	59.7	12344	10677	-6195	53445	54852	D	DHZ	
1996.500	330	12.9	76	57.1	12378	10743	-6149	53411	54827	D	DHZ	
1997.500	330	33.7	76	54.8	12409	10807	-6099	53382	54805	D	DHZ	
1998.500	330	54.7	76	54.2	12416	10850	-6036	53371	54796	D	DHZ	
1999.500	331	17.0	76	51.9	12446	10915	-5980	53336	54769	D	DHZ	
2000.500	331	37.8	76	50.1	12472	10974	-5926	53317	54756	D	DHZ	
2001.500	332	00.3	76	47.0	12512	11048	-5873	53276	54726	D	DHZ	
2002.500	332	23.3	76	45.3	12536	11108	-5810	53256	54711	D	DHZ	
2003.500	332	44.1	76	45.7	12526	11134	-5738	53245	54698	D	DHZ	
2004.500	333	06.5	76	42.6	12567	11208	-5684	53206	54670	D	DHZ	
2005.500	333	29.1	76	40.1	12600	11275	-5625	53174	54647	D	DHZ	
2006.500	333	50.1	76	37.7	12631	11337	-5570	53140	54621	D	DHZ	

* A = All Days
* Q = Quiet Days
* D = Disturbed Days
* I = Incomplete
* J = Jumps jump value = old site value - new site value

ELE = vector variometer sensor orientation

Notes:

1. No data were available from 01 January 1943 until 15 January 1949. From 15 January 1949 until 15 December 1949 there were no data from the vertical (Z) channel.
2. No data available from 2 January until 20 April 1950 due to installation of a new variometer system.
3. The observatory was shut down on 15 December 1951 and re-opened on 19 February 1983
4. The jump in the values in 1989 is due to establishment of a new absolute pillar during 1989
5. A temporary variometer was used from 02 February 1991 until 01 December 1992 when the primary variometer was

repaired after a lightning strike.

6. The jump in the values from 1993 to 1994 is due to a change in the height of the absolute pillar when swapping from QHM and declinometer absolute instruments to DIFlux